

END TERM EXAMINATION

FOURTH SEMESTER [B.TECH] MAY-JUNE 2017

Paper Code: ETMA 202

Subject: Applied Mathematics-IV

Time: 3 Hours

Maximum Marks: 75

Note: Attempt any five questions including Q.No. 1 which is compulsory.
Select one question from each unit.

- Q1. a) Solve $4\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 3u$, given $u = 3e^y - e^{5y}$ when $x=0$, by the method of separation of variables. (3)
- b) Define the skewness and kurtosis. Also write their coefficients. (3)
- c) Determine the Binomial Distribution for which mean=2 (Variance) and mean+ Variance=3. Also find $P(X \leq 3)$. (4)
- d) The first four moments of a distribution about the value '0' are -0.20, 1.76, -2.36 and 10.88. Find the moments about the mean. (4)
- e) State Baye's theorem. (3)
- f) The standard weight of a special purpose brick is 5 kg. and it contains two basic ingredients B_1 and B_2 . B_1 cost Rs. 5 per kg. and B_2 cost Rs. 8 per kg. Strength considerations state that the brick contains not more than 4 kg. of B_1 and minimum of 2 kg of B_2 . Since the demand for the product is likely to be related to the price of the brick, find out graphically minimum cost of the brick satisfying the above conditions. (5)
- g) Find the dual of the following primal problem:
Min. $Z = 3x_1 - 2x_2 + 4x_3$, subject to: $3x_1 + 5x_2 + 4x_3 \geq 7$, $6x_1 + x_2 + 3x_3 \geq 4$, $7x_1 - 2x_2 - x_3 \leq 10$, and $x_1, x_2, x_3 \geq 0$. (3)

Unit-I

- Q2. a) $(D^2 - DD' - 2D'^2)z = (y-1)e^x$. (6.5)
- b) $(D^2 - DD' + D')z = x^2 + y^2$. (6)
- Q3. a) An insulated rod of length l has its end A and B maintained at 0°C and 100°C respectively until steady state conditions prevail. If B is suddenly reduced to 0°C and maintained at 0°C , find the temperature at a distance x from A at time t . (6.5)
- b) Solve the above problem if the change consists of raising the temperature of A to 20°C and reducing that of B to 80°C . (6)

Unit-II

- Q4. a) Three balls are drawn successively from a box containing 6 red balls, 4 white balls, and 5 blue balls. Find the probability that they are drawn in the order red, white and blue if each ball is i) replaced and ii) not replaced. (6.5)
- b) Find i) $E(X)$, ii) $E(X^2)$, iii) $E[(X-X)^2]$ for the probability distribution shown in the following table: (6)

X	8	12	16	202	24
p(X)	1/8	1/6	3/8	1/4	1/12

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Q5. a) A continuous distribution of a variable x in the range $(-3,3)$ is defined as

$$f(x) = \frac{1}{16}(3+x)^2, \quad -3 \leq x < -1,$$

$$= \frac{1}{16}(2-6x^2), \quad -1 \leq x < 1,$$

$$= \frac{1}{16}(3-x)^2, \quad 1 \leq x \leq 3.$$

Verify that the area under the curve is unity. Show that the mean is zero. (6)

b) If the heights of 300 students are normally distributed with mean 68.0 inch and standard deviation 3.0 inch, how many students have heights.
i) greater than 72 inch. ii) between 65 and 71 inch. (6.5)

Unit-III

Q6. a) Fit a second degree parabola to the following data using method of least squares. (6.5)

X	10	12	15	23	20
Y	14	17	23	25	21

b) For 10 observations on price (x) and supply (y), the following data were obtained $\Sigma x=130$, $\Sigma y=220$, $\Sigma x^2=2288$, $\Sigma y^2=5506$, $\Sigma xy=3467$. (6)

Obtain the two lines of regression, correlation coefficient and estimate the supply when the price is 16 units.

Q7. a) A group of 10 rats fed on a diet A and another group of 8 rats fed on a different diet B, recorded the following increase in weight: (6.5)

Diet A (gm.)	5	6	8	1	12	4	3	9	6	10
Diet B (gm.)	2	3	6	8	10	1	2	8	-	-

Does it show that superiority of diet A over that of B.

b) Fit a Poisson distribution to the following data and test for its goodness of fit at 0.05 level of significance. (6)

X:	0	1	2	3	4
F:	419	352	154	56	19

Unit-IV

Q8. Use penalty (or Big-M) method to solve the problem:

Max. $z = 6x_1 + 4x_2$ subject to $2x_1 + 3x_2 \leq 30$, $3x_1 + 2x_2 \leq 24$, $x_1 + x_2 \geq 3$, and $x_1, x_2 \geq 0$.

Is the solution is unique? If not, give two different solutions. (12.5)

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Q9. a) A method Engineer wants to assign four new methods to three work centers. The assignment of the new methods will increase production and they are given below. If only one method can be assigned to a work center, determine the optimum assignment. (6)

Increase in Production (unit)

	Works Centers		
	A	B	C
1	10	7	8
2	8	9	7
3	7	12	6
4	10	10	8

b) Solve the following Transportation problem: (6.5)

Suppliers Consumers	A	B	C	Available
I	6	8	4	14
II	4	9	8	12
III	1	2	6	5
Required	6	10	15	31

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Note: Attempt any five questions including Q.no.1 which is compulsory. Select one question from each Unit. Students can ask for statistical table.

- Q1 (a) Find a particular integral of $(D_x^3 + 3D_x^2 D_y - 2D_x^2)z = (x^2 + 2y)e^{2x+y}$. (7)
 (b) A die is tossed thrice. Getting 2 or 4 on the die in a toss is success. Find the mean and variance of number of success. (6)
 (c) Can $y = 5 + 2.8x$ and $x = 3 - 0.5y$ be the estimated regression equations of y on x and x on y respectively? Explain. (5)
 (d) Write the dual to the following primal problem. (7)

$$\text{Max } Z = 3x_1 + 10x_2 + 2x_3$$

Subject to:

$$2x_1 + 3x_2 + 2x_3 \leq 7$$

$$3x_1 - 2x_2 + 4x_3 = 3$$

Where $x_1, x_2, x_3 \geq 0$.

Prove that dual of the dual is primal.

Unit-I

- Q2 (a) Find the general solution of $(D^3 - 4D^2 D' + 4DD'^2)z = \cos(2x + 3y)$. (6)
 (b) Find the complete solution of the equation: (6.5)
 $(D^2 + D'^2 + 2DD' + 2D + 2D' + 1)z = e^{2x+y}$

- Q3 (a) Solve $\frac{\partial^2 u}{\partial x^2} - 2\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$ where $u(0, y) = 0$ and $\frac{\partial u}{\partial x}(0, y) = e^{-3y}$ for all y using the method of separation of variables. (6)
 (b) A long rectangular plate of width π cm with insulated surfaces has its temperature equal to zero on both the long sides and one of the short side so that $u(0, y) = 0$, $u(\pi, y) = 0$, $u(x, \infty) = 0$ and $u(x, 0) = kx$. Find the steady state temperature within the plate. (6.5)

Unit-II

- Q4 (a) In a bolt factory there are four machines A, B, C and D manufacturing 20%, 15%, 25% and 40% of the total output respectively. Of their outputs 5%, 4%, 3% and 2% in the same order are defective bolts. A bolt is chosen randomly from the factory's production and is found defective. What is the probability that the bolt was manufactured by machine A or D. (6)
 (b) Calculate the first four moments for the following frequency distribution about the mean and explain the skewness and kurtosis of the frequency distribution. (6.5)

X:	-4	-3	-2	-1	0	1	2	3	4
Y:	3	4	5	7	12	7	5	4	3

- Q5 (a) Find mean, variance and moment generating function of $f(x)$, where (6)

$$f(x) = \begin{cases} ae^{-ax}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

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- (b) If the probability that an individual suffers to a bad reaction form an injection of a given serum is 0.001, determine the probability that out of 2000 individuals. (6.5)

(i) exactly 3 (ii) more than 2 individual will suffer to a bad reaction.

Unit-III

- Q6 (a) The two regression equation of the variable x and y are $8x - 10y + 66 = 0$ and $40x - 18y = 214$. Given that variance of x = 9. Find (6)
- (i) the mean of x and y
(ii) the standard deviation of y and
(iii) the coefficient of correlation between x and y.
- (b) The results of measurement of electric resistance R of a copper bar at various temperature $t^{\circ}\text{C}$ are listed below: (6.5)

t:	19	25	30	36	40	45	50
R:	76	77	79	80	82	83	85

If $R = a + bt$, find a and b.

- Q7 (a) Write at least three important properties of Regression coefficient and prove that if two variables are uncorrelated then the regression lines are perpendicular to each other. (6)
- (b) A sample of 10 boxes of chips is drawn in which the mean weight is 490 gm and standard deviation of weight is 9 gm. Can the sample be considered to be taken from a population having mean weight 500 gm where $t_{0.5} = 2.26$? (6.5)

Unit-IV

- Q8 (a) Write the dual of the following problem: (6)

Min. $Z = 2x_1 + 3x_2 + 4x_3$

St. $2x_1 + 3x_2 + 5x_3 = 2$

$3x_1 + x_2 + 7x_3 \leq 3$

$x_1 + 4x_2 + 6x_3 = 5$

where $x_2, x_3 \geq 0$ and x_1 unrestricted.

- (b) Using dual simplex method solve following LPP. (6.5)

Max. $Z = -3x_1 - 2x_2$

Subject to

$x_1 + x_2 \geq 1$

$x_1 + x_2 \leq 7$

$x_1 + 2x_2 \geq 10$

$x_2 \leq 3$

where $x_1, x_2 \geq 0$.

- Q9 Using VAM method find basic feasible solution of the following transportation problem. Check optimality and hence find the optimal solution. (12.5)

From	A	B	C	D	Supply
I	21	16	25	13	11
II	17	18	14	23	13
III	32	27	18	41	19
Demand	6	10	12	15	43

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